Protocol I8R-MC-IGBI(b)

Comparison of Glucagon Administered by Either the Nasal(LY900018) or Intra-muscular (GlucaGen®) Routes in Adult Patients with Type 1 Diabetes Mellitus During Controlled Insulin-Induced Hypoglycemia

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Glucagon nasal powder (LY900018)

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1. Protocol Synopsis

Title of Study:
Comparison of Glucagon Administered by Either the Nasal (LY900018) or Intra-Muscular (GlucaGen®) Routes in Adult Patients with Type 1 Diabetes Mellitus During Controlled Insulin-Induced Hypoglycemia

Rationale:
LY900018 is an investigational drug–device combination product consisting of a single-use nasal dosing device that delivers a fixed 3-mg dose of glucagon powder absorbed through the nasal mucosa. The glucagon component of LY900018 is a synthetic single-chain, 29-amino-acid polypeptide identical to human glucagon. LY900018 is being developed for the treatment of severe hypoglycemia.

Currently, glucagon in a liquid form lacks physical and chemical stability; thus marketed products require reconstitution of glucagon before the product can be administered through subcutaneous or intra-muscular (IM) injection. LY900018 combines stable, synthetic glucagon and a nasal dosing device that for effective use obviates reconstitution, injection, and moreover, patient inhalation. Because of these advantages, LY900018 may offer a significant improvement in the treatment of severe hypoglycemia occurring outside of the hospital setting.

The aim of this study is to compare GlucaGen®, a currently approved product that requires reconstitution prior to administration through IM injection, and LY900018 in those patients who achieve treatment success during controlled insulin-induced hypoglycemia in patients with type 1 diabetes mellitus (T1DM). Treatment success is defined as an increase in plasma glucose (PG) to ≥70 mg/dL (3.9 mmol/L) or an increase of ≥20 mg/dL from the plasma glucose nadir within 30 minutes after receiving glucagon, without the patient receiving additional actions to increase PG. The nadir is defined as the minimum plasma glucose measurement at the time of or within 10 minutes following glucagon administration.

Objectives/Endpoints

Objectives	Endpoints
Primary To compare LY900018 versus GlucaGen® in the percentage of adult patients with T1DM who achieve treatment success during controlled insulin-induced hypoglycemia	The percentage of patients who achieve treatment success will be defined as an increase in plasma glucose to ≥70 mg/dL or an increase of ≥20 mg/dL from plasma glucose nadir within 30 minutes after receiving glucagon, without receiving additional actions to increase the plasma glucose concentration. The nadir will be defined as the minimum plasma glucose concentration at the time of or within 10 minutes following glucagon administration.
Secondary 1. To assess the safety and tolerability of LY900018 versus GlucaGen® 2. To characterize the PD profiles of LY900018 versus GlucaGen® 3. To characterize the PK profiles of LY900018 versus GlucaGen®	1. Summary of AEs, including nasal and non-nasal AEs; vital signs 2. The PD parameters include BGmax, Tmax 3. The PK parameters include AUC, Cmax, Tmax

Abbreviations: AE = adverse event; AUC = area under the concentration versus time curve; BGmax = maximal blood glucose; Cmax = maximal concentration; PD = pharmacodynamics; PK = pharmacokinetics; T1DM = type 1 diabetes mellitus; Tmax = time to maximal concentration.

Summary of Study Design:
Study I8R-MC-IGBI is a multicenter, randomized, open-label, 2-treatment, 2-period, crossover study conducted in adult patients with T1DM.

Treatment Arms and Planned Duration for an Individual Patient:
LY900018: Single 3-mg dose; nasal administration by study site personnel
GlucaGen® (Novo Nordisk A/S): Single 1-mg dose; IM injection

Number of Patients:
Approximately 70 patients with T1DM will be enrolled in order to have 66 patients complete both treatment visits with evaluable primary outcome.

Statistical Analysis: A total of 66 patients are required in the study in order to achieve the primary objective with at least 90% power using the following assumptions:

- A treatment success rate of 98% for both treatments
- A noninferiority margin of 10%
- One-sided alpha level of 0.025
- A within-patient correlation of zero between 2 treatment visits

The primary analysis will be a treatment comparison of the percentage of patients who achieve treatment success. The percentage of patients who achieve treatment success within each treatment arm and the difference in the percentages between 2 treatment arms will be computed. A 2-sided 95% confidence interval (CI) will be obtained from the 1-sample mean of the paired differences in outcome (1=outcome observed; 0=outcome not observed) across 2 study visits. Noninferiority of nasal glucagon (NG) will be declared if the upper limit of the 2-sided 95% CI constructed on the difference in percentage (IM glucagon – NG) is less than the noninferiority margin of 10%.

2. Schedule of Activities

Study Schedule Protocol I8R-MC-IGBI

Procedure	Screening Visit	Period 1 and Period 2 CRU Visits	Follow-up (or ED) Visit	Comments
	Up to 28 days before Period 1	Day 1	28(±2) days after last dose of study drug	
Sign informed consent form	X			Written informed consent has to be provided before any trial-related activity and may be obtained during a separate informed consent visit to the site
Medical history	X			
Clarke Hypoglycemia Awareness Survey	X			See Appendix 8.
Height	X			
Weight	X	X	X	During Periods 1 and 2, body weight will be collected before the start of the procedure to induce hypoglycemia
Vital signs (supine blood pressure, pulse rate, and body temperature)	X	X	X	See Section 9.5.2; the patient should be at rest for at least 5 minutes before vital signs are measured. During Periods 1 and 2, vital signs should be measured before the start of the procedure to induce hypoglycemia and at 45 minutes after glucagon administration. Time points may be added for Period 1 and Period 2 if warranted and agreed upon between Lilly and the investigator.
Clinical laboratory tests	X	X	X	See Appendix 2, Clinical Laboratory Tests, for details. Patients do not need to fast for laboratory samples collected at screening or follow-up. At Period 1 and Period 2 site visits, laboratory tests should be collected from patients who have fasted at least 8 hours before any study procedures. All screening, hepatic monitoring and study day laboratory tests to be processed at a local laboratory. Hepatitis E laboratory tests to be processed at a central laboratory.
Sample collection for HbA1c	X	X (Period 1 only)		Screening samples and samples collected before the start of the procedure to induce hypoglycemia to be processed at local laboratory.

Pregnancy test	X	X	X	Serum pregnancy test will be performed at screening and analyzed at a local laboratory. Urine pregnancy test will be performed at admission for study Periods 1 and 2 as well as at the follow-up visit and processed on site
Physical examination	X	X	X	Physical examinations will be performed before the start of the procedure to induce hypoglycemia in Periods 1 and 2 and at the follow-up visit. Patients will review recent insulin demand with site personnel prior to inducing hypoglycemia (see Section 7.7).
12-Lead ECG	X	X	X	See Section 9.5.3; single 12-lead ECGs will be collected for local safety assessment. During Periods 1 and 2, ECGs will be collected before the start of the procedure to induce hypoglycemia. Electrocardiograms may be obtained at additional times, when deemed clinically necessary
Adverse event assessment	X	X	X	See Section 9.3
Admission to CRU		X		The patient should arrive at the CRU at approximately 0700 hours, having fasted for at least 8 hours
Discharge from CRU		X		The patient may be discharged 6 hours following the procedure. A patient may remain inpatient at the CRU at the discretion of the investigator
Dinner		X		Following completion of all study procedures, the patient will be provided a carbohydrate-rich meal, and the investigator will ensure the patient's plasma glucose is stable
Nasal inspection		X		Nasal inspection should be conducted before the procedure to induce hypoglycemia and at 90 minutes postglucagon administration
Injection-site inspection (for IM glucagon only)		X		The injection site will be inspected before the procedure to induce hypoglycemia to confirm injection site is normal and at 90 minutes postglucagon administration
Nasal and Non-nasal Score Questionnaire		X		See Appendix 6, Nasal and Non-nasal Score Questionnaire. Questionnaire to be completed approximately 30 minutes prior to hypoglycemia induction and 15, 30, 60, and 90 minutes postglucagon administration

Edinburgh Hypoglycemia Scale		X		See Appendix 7, Edinburgh Hypoglycemia Scale.This questionnaire should be completed when PG ≤75 mg/dL and again at PG ≤60 mg/dL after the start of the insulin infusion and at 15, 30, 45, and 60 minutes postglucagon administration
Insulin infusion to induce hypoglycemia		X		Refer to Sections 9.2 and 7.7 for details regarding this procedure, including steps to take before the procedure begins and steps to take if there is an insufficient response to glucagon administration. Procedures taking place on Period 2 will occur at approximately the same time as those for Period 1.
Study drug administration		X		Once PG reaches below 60 mg/dL, the insulin infusion is stopped and approximately 5 minutes later study drug is administered
Sample collection for measurement of glucagon		X	X	Samples will be collected pre-dose (i.e., immediately prior to glucagon administration) and at 5, 10, 15, 20, 25, 30, 40, 50, 60, 90, 120, and 240 minutes post-glucagon administration at Periods 1 and 2 and once at follow-up. All samples will be processed at a central laboratory.
Sample collection for measurement of DPC		X		Samples will be collected from NG-treated patients pre-dose (i.e., immediately prior to glucagon administration) and at 10, 20, 30, 40, 60, 120, and 240 minutes post-glucagon administration. Three additional samples may be collected. Samples to be processed at a central laboratory.
Bedside glucose safety measurements (SuperGL laboratory analyzer)		X		Samples will be collected throughout the study procedure. During the hypoglycemia induction procedure, plasma glucose concentration will be measured no more than 10 minutes apart while the plasma glucose concentration is ≥100 mg/dL and no more than 5 minutes apart when the plasma glucose concentration is <100 mg/dL. Post-glucagon administration, plasma glucose concentration will be measured every 5 minutes for the first 30 minutes and every 10 minutes up to 90 minutes.
Sample collection for measurement of plasma glucose		X		Samples will be collected pre-dose (i.e., immediately prior to glucagon administration) and at 5, 10, 15, 20, 25, 30, 40, 50, 60, and 90 minutes post-glucagon administration. Samples to be processed at a central laboratory.
Exploratory sample		X (Period 1 only)	X	Serum samples will be collected prior to the procedure to induce hypoglycemia.

Genetic sample		X (Period 1 only)		

Abbreviations: CRU = clinical research unit; DPC = dodecylphosphocholine; ECG = electrocardiogram; ED = early discontinuation; HbA1c = hemoglobin A1c; IM = intra-muscular; NG = nasal glucagon; PG = plasma glucose.

Note: if multiple procedures take place at the same time point, the following order of the procedure should be used: ECG, vital signs and venipuncture.

3. Introduction

3.1. Study Rationale

Nasal glucagon (NG; LY900018) is an investigational drug–device combination product consisting of a single-use nasal dosing device that delivers glucagon powder for absorption through the nasal mucosa. Nasal glucagon represents a significant improvement in the evolution of the treatment of severe hypoglycemia with glucagon; it provides the same single-chain, 29-amino-acid polypeptide as human glucagon and the recombinant DNA-produced glucagon used in currently marketed glucagon emergency kits (Glucagon Emergency Kit [Lilly] and GlucaGen® HypoKit [Novo Nordisk A/S]).

The aim of this study (Study I8R-MC-IGBI [IGBI]) is to compare the planned commercial drug product of NG to GlucaGen, a currently approved product that requires reconstitution prior to administration through intra-muscular (IM) injection, in those patients who achieve treatment success during controlled insulin-induced hypoglycemia in patients with type 1 diabetes mellitus (T1DM). Treatment success is defined as an increase in plasma glucose (PG) concentration to ≥70 mg/dL or an increase of ≥20 mg/dL from the PG nadir within 30 minutes after receiving glucagon, without the patient receiving additional actions to increase PG. The nadir is defined as the minimum PG concentration at the time of or within 10 minutes following glucagon administration. The design and conduct of Study IGBI are similar to the completed Phase 3 Study I8R-MC-IGBC (IGBC) (Rickels et al. 2016).

3.2. Background

Glucagon is a widely available product currently indicated for the treatment of severe hypoglycemia. The principal action of glucagon is to increase PG concentration through its action in the liver. Nasal glucagon development has produced robust clinical data in adult healthy subjects, adult patients with T1DM and type 2 diabetes mellitus (T2DM), as well as pediatric patients with T1DM.

Nasal glucagon powder contains 3-mg of glucagon in a ready-to-use, single-use intranasal powder delivery device. Glucagon comprises 10% by weight of the powder in the device. The formulation also contains beta-cyclodextrin as a filler/bulking agent/absorption enhancer and dodecylphosphocholine (DPC) as an absorption enhancer/surfactant. Beta-cyclodextrin is a compendial excipient and DPC is a novel excipient. The device delivers a single dose to the nostril upon actuation.

Three clinical trials using NG have been completed in adults with T1DM and T2DM: Study IGBC, Study I8R-MC-IGBA (IGBA), and an actual use study (Study I8R-MC-B002 [B002]). Studies IGBC and IGBA demonstrated comparable safety and efficacy between NG and injectable glucagon in reversing insulin-induced hypoglycemia in adult patients with T1DM only (Study IGBA) or patients with T2DM and T1DM (IGBC). Study B002 was an actual use study that evaluated the effectiveness of NG administered by a trained caregiver to patients with T1DM experiencing moderate to severe hypoglycemia in a real-world environment of work and home. Study B002 demonstrated that 96% of moderate to severe hypoglycemic events were

resolved within 30 minutes. Finally, 2 trials were conducted in pediatric patients with T1DM: Study I8R-MC-IGBB (IGBB) and Study I8R-MC-B001 (B001). Studies IGBB and B001demonstrated effectiveness in rescuing pediatric patients from hypoglycemia.

All adult diabetic patients in the 2 inpatient clinical trials (Studies IGBA and IGBC) underwent hypoglycemia induction through intravenous (IV) insulin under close clinical supervision and were administered either NG or injectable glucagon.

Patients fully recovered from hypoglycemia without additional actions to increase glucose level. Specifically, in Study IGBC, at 30 minutes after glucagon dosing, 98.7% (74 out of 75 patients) of NG and 100% (75 out of 75) of intra-muscular glucagon (IMG)-treated patients achieved treatment success (defined as an increase in PG to $\geq$70 mg/dL or an increase of $\geq$20 mg/dL from nadir). The nadir is defined as the minimum glucose measurement at the time of or within 10 minutes following glucagon administration. The NG-treated patients who did not meet the above criteria did achieve both PG $\geq$70 mg/dL and an increase of $\geq$20 mg/dL from nadir at 40 minutes after dosing. Furthermore, the mean time to treatment success was 16 minutes in the NG treatment arm and 13 minutes in the IMG treatment arm (Rickels et al. 2016).

3.3. Benefit–Risk Assessment

Glucagon is generally well tolerated, with mild side effects noted. These included occasional nausea and vomiting; however, these side effects may also occur with hypoglycemia (for example, dizziness, fainting, light headedness, blurred vision, headaches, fatigue). Nasal glucagon is a needle-free, ready-to-use nasal dosing device for administration into nostrils requiring no inhalation. In contrast to other products on the market, this requires no reconstitution. Patients being treated with the investigational product will derive no direct benefit from it as treatment will follow an insulin-induced hypoglycemia.

To date, there have been 10 completed clinical studies (8 in adults, 2 in pediatric population) in which the safety, efficacy, or effectiveness of NG has been evaluated. No deaths or serious adverse events (SAEs) associated with NG occurred during these studies. The most frequently reported treatment-emergent adverse events (TEAEs) in NG-treated patients represent events related to local tolerability including lacrimation increased, headache, rhinorrhea, nasal discomfort, nasal congestion, and nasal pruritus, and gastrointestinal disorders including nausea and vomiting. All TEAEs were transient in nature and most resolved within 24 hours of exposure. Severe adverse reactions to glucagon are very rare, although generalized allergic reactions, urticaria, respiratory distress, and hypotension have been reported in patients who have received glucagon.

This study will expose patients with T1DM to an insulin-induced hypoglycemia meant to simulate hypoglycemia in a controlled setting. The protocols used to induce the controlled hypoglycemia have been adopted from a previously completed trial, specifically Study IGBC, that demonstrated the safe use of this method. This is an inpatient procedure in which the patients will be under constant supervision of the clinical research unit (CRU) staff. Safety provisions have been considered in that IV glucose will be administered if the patient

experiences signs and symptoms of severe hypoglycemia and gauged to require intervention during the experimental procedure, at the discretion of the investigator.

No clinically significant safety or tolerability concerns have been identified in patients to date for NG up to the highest single dose (6-mg) given in an acute setting.

More information about the known and expected benefits, risks, SAEs, and reasonably anticipated adverse events (AEs) of glucagon nasal powder are to be found in the Investigator's Brochure (IB).

4. Objectives and Endpoints

Table IGBI.1 shows the objectives and endpoints of the study.

Table IGBI.1. Objectives and Endpoints

Objectives	Endpoints
Primary To compare LY900018 versus GlucaGen® in the percentage of adult patients with T1DM who achieve treatment success during controlled insulin-induced hypoglycemia	The percentage of patients who achieve treatment success, which will be defined as an increase in plasma glucose to ≥70 mg/dL or an increase of ≥20 mg/dL from plasma glucose nadir within 30 minutes after receiving glucagon, without receiving additional actions to increase the plasma glucose concentration; nadir will be defined as the minimum plasma glucose concentration at the time of or within 10 minutes following glucagon administration.
Secondary 1. To assess the safety and tolerability of LY900018 versus GlucaGen® 2. To characterize the PD profiles of LY900018 versus GlucaGen® 3. To characterize the PK profiles of LY900018 versus GlucaGen®	1. Summary of AEs, including nasal and non-nasal AEs; vital signs 2. The PD parameters include BGmax, Tmax 3. The PK parameters include AUC, Cmax, Tmax
Exploratory 1. To characterize the PK profile of the DPC excipient of LY900018 2. To assess the occurrence and severity of hypoglycemia symptoms during controlled insulin-induced hypoglycemia 3. To assess the awareness of hypoglycemia prior to its induction	1. The PK parameters include AUC, Cmax, Tmax 2. The Edinburgh Hypoglycemic Scale will be used to assess the occurrence and severity of hypoglycemia symptoms 3. The Clarke Hypoglycemia Awareness survey will be used to assess awareness of hypoglycemia symptoms

Abbreviations: AE = adverse event; AUC = area under the concentration versus time curve; BGmax = maximal blood glucose; Cmax = maximal concentration; DPC = dodecylphosphocholine; PD = pharmacodynamics; PK = pharmacokinetics; T1DM = type 1 diabetes mellitus; Tmax = time to maximal concentration.

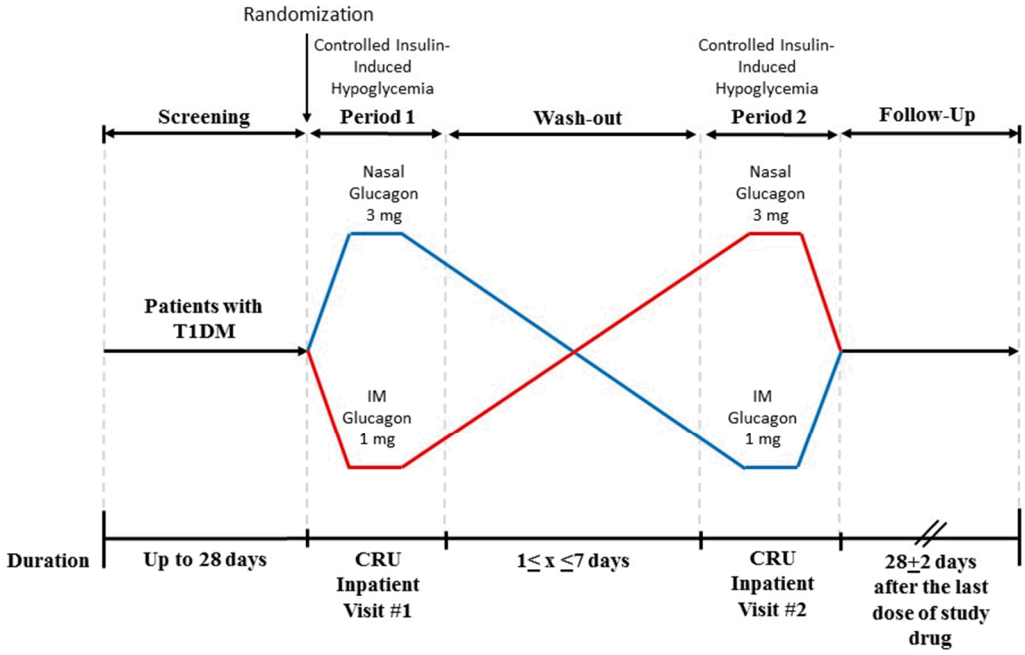
5. Study Design

5.1. Overall Design

This is a Phase 1, multicenter, randomized, open-label, 2-treatment, 2-period, crossover study. Figure IGBI.1 illustrates the study design. In each period, patients with T1DM will undergo a procedure to induce hypoglycemia using insulin infusion. The insulin infusion will be used to decrease PG and will be stopped once PG <60 mg/dL and it is expected that PG will continue to decrease for an additional 15 to 20 mg/dL to its lowest concentration. Approximately 5 minutes after the insulin infusion is stopped, patients will be administered either 3-mg NG (LY900018) or 1-mg IMG (GlucaGen) in the deltoid muscle of the nondominant arm by CRU staff members. Serial blood sampling will be performed for glucagon and PG concentration assessments immediately before and up to 6 hours following administration of glucagon. Following each test period, patients will remain in the CRU for at least 6 hours following the procedure. All patients will receive a carbohydrate-rich meal prior to discharge. Patients may stay longer as needed, at the discretion of the investigator.

After a wash-out period of at least 1 day (24 hours) and no more than 7 days, patients will return to the clinic and the procedure repeated with each patient crossed over to the alternate glucagon treatment.

A CRU visit for follow-up or early discontinuation should occur approximately 28±2 days after the last dose of study drug.



Abbreviations: CRU = clinical research unit; IM = intra-muscular; T1DM = type 1 diabetes mellitus.

Figure IGBI.1. Illustration of study design for Protocol I8R-MC-IGBI.

5.2. Number of Participants

Approximately 70 patients with T1DM may be enrolled to ensure that at least 66 patients complete both treatment visits with evaluable primary outcome (see Section 10.3.1.1).

If patients drop out of the study before completion of both periods or completed the study without evaluable primary outcome in at least 1 period, replacement patients will be enrolled to ensure at least 66 patients are available for the evaluation of primary objective. The replacement patients will assume the same treatment sequence as the patients to be replaced and will complete that treatment sequence in its entirety.

5.3. End of Study Definition

End of the study is the date of the last visit or last scheduled procedure shown in the Schedule of Activities (Section 2) for the last patient.

5.4. Scientific Rationale for Study Design

This study will be used to supplement the comprehensive chemistry, manufacturing control analytical plan for bridging the planned commercial drug product with the clinical trial drug product. Since the existing stores of clinical trial drug product have been exhausted, an indirect comparator/bridging study must be performed to demonstrate efficacy comparability.

This study design involves an open-label assessment of the planned commercial drug product compared to the marketed product (GlucaGen) within the T1DM population. The 2 study periods will be separated by a washout period of no less than 1 day (24 hours) and no more than 7 days due to the short half-life of NG (approximately $t_{1/2}$ = 25 minutes) The second of the 2 study periods will expose the same patients to the alternate study arm (for example, those randomly assigned to receive NG LY900018 will receive IM GlucaGen for Period 2). This crossover design will allow each individual patient to act as an internal control.

5.5. Justification for Dose

Results from studies in both pediatric and adult populations confirm that, while the physiological response to glucagon does appear to saturate between 2- and 3-mg NG dose, the lower 2-mg dose may not always elicit the maximum response needed in an emergency situation of severe hypoglycemia. The 3-mg dose provides more consistent clinical efficacy compared to the 2-mg dose and is similarly well tolerated in both pediatric and adult patients.

Based on LY900018 exposure, efficacy, and safety, the 3-mg dose was selected for the Phase 3 studies. Therefore, in the current study, the 3-mg NG dose will be used to compare against 1 mg of IM GlucaGen.

6. Study Population

Eligibility of patients for the study will be based on the results of screening medical history, physical examination, vital signs, clinical laboratory tests, and electrocardiogram (ECG). Refer Section 6.1 and Section 6.2 for full inclusion/exclusion criteria.

The nature of any conditions present at the time of the physical examination and any preexisting conditions will be documented.

Screening may occur up to 28 days prior to enrollment. Patients who are not enrolled within 28 days of screening may be subjected to an additional medical assessment and/or clinical measurements to confirm their eligibility.

Prospective approval of protocol deviations to recruitment and enrollment criteria, also known as protocol waivers or exemptions, is not permitted.

6.1. Inclusion Criteria

Patients are eligible for inclusion in the study only if they meet all of the following criteria at screening and/or enrollment:

[1] are male or female with a diagnosis of T1DM for at least 2 years and receiving daily insulin since the time of diagnosis

[1a] for sexually active male patients, who have not been surgically sterilized, with a female partner of childbearing potential, the female partner must use a highly effective method of contraception (for example, surgical sterilization, hormonal intrauterine devices [coil], or oral hormone contraceptives, each in combination with spermicide-coated condoms); when female partners do not use these methods, the male patient must be willing to refrain from sexual intercourse from the first dosing until 1 month after the last dosing in the trial

[1b] female patients of child-bearing potential must agree to use a highly effective method of birth control (highly effective contraception methods are considered those with a failure rate <1% undesired pregnancies per year and include surgical sterilization, hormonal intrauterine devices [coil], oral hormone contraceptives, sexual abstinence, or a surgically sterilized partner) throughout the entire duration of this study (from the screening visit until study completion)

females of nonchildbearing potential may participate without using highly effective contraceptive methods which include those who are postmenopausal, defined as women aged <52 years and being amenorrheic for more than 1 year with a serum follicle-stimulating hormone (FSH) level compatible with postmenopausal status or aged ≥52 years and being amenorrheic for less than 1 year and with serum FSH level according to local laboratory reference range and not using highly effective contraceptive methods or aged ≥52 years being amenorrheic for more than 1 year.

[2] are between the ages of 18 to 64 years, inclusive

[3] have a body weight of at least 50 kg (110 lb)

[4] have a body mass index of 18.5 to 35 kg/m^2, inclusive

[5] have a hemoglobin A1c (HbA1c) value ≤10%

[6] have clinical laboratory test results within normal reference range for the population or investigational site, or results with acceptable deviations that are judged to be not clinically significant by the investigator

[7] have venous access sufficient to allow for blood sampling and required infusions as per the protocol

[8] are reliable and willing to make themselves available for the duration of the study and are willing to follow study procedures

[9] has provided signed informed consent

6.2. Exclusion Criteria

Patients will be excluded from study enrollment if they meet any of the following criteria at screening and/or enrollment:

[10] are investigative site personnel directly affiliated with this study and their immediate families. Immediate family is defined as a spouse, biological or legal guardian, child, or sibling

[11] are Lilly employees

[12] are currently enrolled in a clinical study involving an investigational product or any other type of medical research judged not to be scientifically or medically compatible with this study

[13] have participated, within the past 30 days, in a clinical study involving an investigational product

[14] have previously completed or withdrawn from this study or any other study investigating LY900018, and have previously received the investigational product

[15] have known allergies to LY900018, related compounds or any components of the formulation, or history of significant atopy

[16] have a history of hypersensitivity to glucagon or any related products or severe hypersensitivity reactions (such as angioedema) to any drugs

[17] have an abnormality in the 12-lead ECG that, in the opinion of the investigator, increases the risks associated with participating in the study

[18] have a supine blood pressure at screening (after at least 5 minutes at rest) outside the range of 95 to 140 mmHg systolic blood pressure and/or greater than 90 mmHg for diastolic blood pressure (this excludes white-coat hypertension, and if repeated measurements are within this range, the subject may be included in the study); have symptoms of arterial hypotension and/or a heart rate at rest outside the range of 50 to 90 beats per minute

[19] have a history of pheochromocytoma (that is, adrenal gland tumor) or insulinoma

[20] occurrence of an episode of severe hypoglycemia (defined as requiring the assistance of another person in the 1 month prior to enrolling in the study)

[21] have a history of epilepsy or seizure disorder

[22] have a significant history of or current cardiovascular, respiratory, hepatic, renal, gastrointestinal, endocrine (apart from T1DM), hematological, or neurological disorders capable of significantly altering the absorption, metabolism, or elimination of drugs; of constituting a risk when taking the study drug; or of interfering with the interpretation of data

[23] have a significant history of or current psychiatric disorders.

[24] regularly use known drugs of abuse and/or show positive findings on drug screening

[25] show evidence of human immunodeficiency virus (HIV) infection and/or positive human HIV antibodies

[26] show evidence of hepatitis C and/or positive hepatitis C antibody

[27] show evidence of hepatitis B and/or positive hepatitis B surface antigen

[28] are women who are pregnant according to a positive serum pregnancy test, actively trying to get pregnant, or are lactating

[29] have, except for the current regimen of insulin therapy and concomitant medication (for example, antihypertensive agents, lipid-lowering agents, contraceptives, and stable treatment with hormone replacement and thyroid hormone replacement therapy), regular use of or intended use of any over-the-counter or prescription medications or nutritional supplements that treat hyperglycemia or insulin resistance or that promote weight loss within 14 days before dosing

[30] daily use of systemic beta-blocker, indomethacin, warfarin or anticholinergic drugs.

[31] have had blood donation or blood loss of more than 500 mL within the past 3 months

[32] have a significant history of alcoholism or drug abuse as judged by the investigator, or regularly consume more than 24 g alcohol/day (males) or 12 g alcohol/day (females)

[33] in the opinion of the investigator or sponsor, are unsuitable for inclusion in the study

[34] any significant changes in insulin regimen and/or unstable blood glucose control within the past 3 months prior to screening as assessed by the investigator

[35] require daily insulin treatment >1.5 U/kg/body weight.

6.3. Lifestyle and/or Dietary Requirements

Throughout the study, patients may undergo medical assessments and review of compliance with requirements before continuing in the study. Patients should continue to meet the restrictions related to alcohol, caffeine, and medication use.

6.3.1. Meals and Dietary Restrictions

In both treatment periods, patients will fast for at least 8 hours prior to beginning the hypoglycemia induction procedure. Once this procedure is completed, the patients will receive a carbohydrate-rich meal. Patients will be given access to calorie-free water during the hypoglycemic procedure.

While resident in the CRU, patients may not consume any food or caloric drinks other than that provided by the CRU. When not resident in the CRU, patients will be encouraged to follow their normal diets. However, patients are encouraged to consume at least 100 g of carbohydrates during the washout period.

6.3.2. Alcohol

Please refer to Section 6.2 regarding excessive alcohol use. Alcohol use will be prohibited during the fasting procedure. However, alcohol may be consumed during the washout period as long as it conforms with the exclusion criteria and does not deviate from habitual patterns of consumption.

6.3.3. Activity

Patients are encouraged to maintain their regular exercise habits for the duration of the study. However, patients should avoid strenuous exercise 48 hours prior to study days.

During the procedure to induce hypoglycemia, patients should remain recumbent or sitting in the CRU until the end of the procedure.

6.4. Screen Failures

Individuals who do not meet the criteria for participation in this study (screen failure) may not be re-screened.